Investigating the Interaction Between SAR Normalization and Data Augmentation for Cross-Regional Generalization in Sentinel-1 Water Segmentation

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Abstract

Deep learning models for Synthetic Aperture Radar (SAR) surface-water segmentation often lose accuracy on geographically unseen flood events. Normalization and augmentation are routine steps, yet their interaction is poorly understood. We examine how four SAR normalization strategies (unscaled raw decibels, Min-Max scaling, Z-score standardization and percentile clipping with standardization) interact with geometric and radiometric intensity augmentations across ten U-Net experiments, comprising eight controlled experiments and two exploratory pilot runs, on 446 hand-labeled Sentinel-1 chips from Sen1Floods11. Models are trained on five source regions, validated on three and tested on held-out in-distribution (ID) chips plus three unseen out-of-distribution (OOD) events (Bolivia, Mekong, Spain). Across the eight controlled experiments, retaining native decibel values without augmentation gives the highest OOD Intersection over Union (IoU, 0.7837), ahead of Z-score standardization (0.7771), percentile clipping with standardization (0.7760) and Min-Max scaling (0.6889). Each controlled augmentation or filtering configuration falls below its corresponding unaugmented normalization baseline in overall OOD IoU: intensity 0.7614, combined 0.7311 and Lee-style filtering 0.7447 versus 0.7771 (Z-score), and percentile-with-intensity 0.7538 versus 0.7760. The only fully controlled interaction, the difference-in-differences contrast of percentile clipping with intensity augmentation, is +7.24 points in Bolivia but -0.65 overall, so clipping redistributes rather than improves the augmentation effect. Two pilot runs with 43 extra chips suggest a geometric crossover (Min-Max +5.11, Z-score -2.41 points), a hypothesis pending retraining on an identical split. Backscatter statistics are consistent with accuracy depending on water-to-land contrast and dark-land overlap, but only descriptively. All runs are single-seed, so no statistical significance is claimed.

Keywords: Synthetic Aperture Radar, Flood Segmentation, Domain Generalization, Data Normalization, U-Net

1. Introduction

Accurate delineation of surface-water extent during flood events is essential for emergency response, damage assessment and hydrological modeling [1], [2]. Optical satellite sensors such as Sentinel-2 and Landsat-8 provide high-resolution multispectral imagery, but persistent cloud cover, heavy precipitation and solar illumination requirements restrict their availability during active flooding [1], [9]. Synthetic Aperture Radar (SAR) sensors, particularly the European Space Agency's C-band (5.405 GHz) Sentinel-1 mission, avoid these constraints by transmitting microwave pulses that penetrate clouds and operate day and night [1], [2]. Calm open water acts as a specular reflector that directs radar energy away from the side-looking sensor, producing low backscatter returns, whereas rough terrain, vegetation and urban structures scatter energy back toward the sensor and give higher backscatter values [2], [9].

Despite this physical contrast, automated SAR flood segmentation is complicated by coherent speckle noise, variations in incidence angle, wind-roughened water surfaces and topographic shadow or layover effects [2], [8]. While Convolutional Neural Networks (CNNs) such as U-Net [3] outperform classical Otsu histogram thresholding [10] on benchmark datasets like Sen1Floods11 [1], models evaluated on random image splits suffer substantial accuracy drops when tested on geographically unseen flood events [4], [5], [6]. Differences in soil moisture, vegetation canopy structure and terrain slope across hydro-climatic regimes cause models to overfit the backscatter distributions of their source domains [4], [6].

In existing SAR flood-mapping studies, preprocessing choices such as decibel (dB) clipping, Min-Max scaling, Z-score standardization, percentile clipping and Lee-style log-domain speckle filtering [8] are typically combined heuristically with spatial and radiometric augmentations [2], [7], [11]. It remains unclear whether normalization and augmentation act additively or interact nonlinearly across distinct geographical environments. Rotating or flipping side-looking SAR imagery, for example, changes the orientation of radar shadows in pixel coordinates relative to the sensor look direction, while synthetic intensity perturbations can push unclipped radar outliers into unrealistic numerical ranges.

This study evaluates the interaction between SAR normalization strategies and data augmentation regimes on the 446 hand-labeled Sentinel-1 chips of the Sen1Floods11 benchmark [1]. To isolate preprocessing effects, the U-Net architecture, training budget, masked loss function and geographical data splits are held constant across ten selected experimental configurations (E0 through E8, plus a factorial Min-Max geometric variant). Because an exhaustive 16-cell factorial grid was too costly to run, we evaluate a targeted ten-experiment matrix that isolates the main normalization effects, the main augmentation effects and selected interaction terms. Nine of the sixteen normalization–augmentation combinations were evaluated, together with one separate Lee-style filtering experiment (E8) that is not a cell of that factorial, leaving seven combinations untested. Eight of the ten runs share an identical 256-chip training split and are treated as controlled comparisons. The remaining two runs (E4_ZScore_Geometric and MinMax_Geometric) were exploratory pilot experiments trained on 43 additional chips before the in-distribution holdout was iso-

lated. The term interaction is used here in its statistical sense, as a non-additive dependence of the augmentation effect on the normalization applied; only one such cell (percentile clipping $\times$ intensity) is estimated under fully matched training conditions, and the geometric crossover suggested by the pilot runs is treated throughout as a hypothesis, not as an established result. This work makes four contributions:

1. A controlled comparison of four SAR input representations (raw dB, Min-Max scaling, Z-score standardization and 1st–99th percentile clipping with standardization), evaluated on both familiar in-distribution (ID) and unseen out-of-distribution (OOD) flood events.
2. An isolated evaluation of radiometric intensity and combined augmentations, plus a 5×5 Lee-style log-domain speckle filtering baseline [8], on cross-regional generalization.
3. A controlled comparison of nonzero- versus zero-centered normalization under identical augmentation (percentile clipping scores +9.23 percentage points of Bolivia IoU above unclipped Z-score intensity augmentation, a direct configuration difference whose interaction contrast is reported in Section 4.2), plus an explicitly untested hypothesis of a Min-Max $\times$ geometric crossover observed only in the two pilot runs.
4. A quantitative backscatter analysis across all 11 Sen1Floods11 events that associates regional segmentation performance with VV/VH water-to-land contrast (Δ dB) and dark-land histogram overlap below -15 dB, reported as a descriptive association rather than a demonstrated cause.

To ensure full reproducibility, the complete source code, model configurations, evaluation split files and trained U-Net weights are publicly available at https://github.com/Sandeepssh/FloodShift.

2. Related Work

2.1 Deep Learning for SAR Flood Segmentation

Early automated SAR flood mapping relied on radiometric thresholding algorithms, such as global Otsu thresholding [10] and split-based bimodal histogram tiling [9], to separate low-backscatter water from higher-backscatter land. Single-threshold methods, however, struggle when wind roughens water surfaces or when emergent vegetation increases backscatter through double-bounce reflections [2]. Following the introduction of Fully Convolutional Networks (FCNs) [12], encoder-decoder architectures such as U-Net [3], ResNet-backed segmentation networks [13], DeepLabv3+ [14] and UNet++ [15] became the standard for pixel-level flood delineation [1], [11], [16]. Bonafilia et al. [1] introduced the globally distributed Sen1Floods11 benchmark, demonstrating that FCNs and U-Net models outperform a standard -15 dB VV backscatter threshold on both permanent-water and flood-water segmentation. Subsequent studies have combined Sentinel-1 and Sentinel-2 imagery [16] or paired SAR intensity with interferometric coherence in urban and agricultural settings [17], [18] to improve flood detection, although single-date dual-polarization (VV/VH) segmentation is still needed for rapid emergency response when pre-flood baseline acquisitions are unavailable [1], [7].

2.2 SAR Preprocessing and Speckle Filtering

Because raw SAR intensity follows a multiplicative exponential distribution with heavy right-skewness, standard pipelines convert linear power values into logarithmic decibels (dB) [1], [2]. To suppress the granular speckle noise caused by coherent wave interference, classical adaptive spatial filters, such as the Lee-style log-domain local-statistics filter used here [8], compute local window statistics to smooth workflows, however, there is no consensus on input normalization. Some studies apply fixed physical Min-Max clipping (e.g., $[-30, 0]$ dB scaled to $[0, 1]$) [11], others apply global Z-score standardization [1], [4], and recent frameworks use percentile Winsorization (e.g., 1st–99th percentile clipping) prior to standardization to suppress extreme double-bounce urban spikes [7]. No study has systematically isolated whether convolutional BatchNormalization layers [19] make external scaling redundant, or whether non-zero-centered scaling interacts adversely with spatial augmentations.

2.3 Domain Generalization Across Flood Events

Recent literature shows that random chip-level train-test splits overestimate model accuracy because of spatial autocorrelation among tiles from the same flood event [4], [5], [6]. Turkmenli et al. [4] evaluated domain generalization methods across individual SAR flood events treated as distinct source domains, reporting IoU gains of roughly 3 percentage points on unseen events. Dhagat [5] compared random splits against country-level geographical holdouts on Sentinel-1 and Sentinel-2 imagery, showing that a vanilla U-Net IoU drops from 0.727 under random splitting to 0.464 under country-level holdouts. Similarly, Ratté-Fortin et al. [6] investigated leave-one-location-out spatial transferability across 18 flood events, with attention to the role of hydro-climatic and geomorphological similarity between training and target regions. Adhikari et al. [7] combined percentile-clipped VV/VH inputs with Joint Research Centre (JRC) permanent-water priors in a UNet++ framework evaluated on held-out events in Bolivia and Nepal. While these studies confirm the severity of geographic domain shift, they hold preprocessing and augmentation fixed; the present work isolates how normalization and augmentation choices interact across held-out regions.

3. Methodology

3.1 Dataset and Geographical Splitting

Experiments use the hand-labeled Sentinel-1 subset (S1Hand and LabelHand) of the Sen1Floods11 (v1.1) dataset [1]. Although Sen1Floods11 includes 4,831 weakly labeled chips derived from optical/radar thresholding, those weak labels contain classification noise that can distort cross-regional evaluation [1]. We therefore restrict all training, validation and testing to the 446 expert hand-labeled chips. Each chip is a 512×512 pixel dual-polarization SAR image (two channels: VV, vertical transmit/vertical receive, and VH, vertical transmit/horizontal receive) at 10 m spatial resolution, paired one-to-one with a ground-truth mask. Pixel intensities are provided in calibrated logarithmic decibels (dB). Ground-truth masks encode three integer values: 1 for surface water (floodwater

and permanent water), 0 for non-water land, and -1 for invalid pixels along satellite swath borders. Validity is therefore taken from the label mask plus a finiteness check on VV/VH (Eq. 1): the rasterio NoData metadata (src.nodata) and src.dataset_mask() were not consulted separately, and no finite NoData sentinel value was searched for, so a raster that encoded NoData by any means other than the label -1 or a non-finite backscatter value would not have been excluded by Eq. (1). The pixels this could affect, and the checks it motivates, are listed in Section 5.4.

To evaluate both familiar-region performance and unseen cross-regional transferability [5], the 11 flood events are partitioned into four mutually exclusive subsets:

1. **Training set (familiar regions, 256 chips):** sampled from five source events: USA (59), India (58), Paraguay (57), Ghana (46) and Sri Lanka (36).
2. **In-distribution (ID) test set (familiar regions, 43 chips):** held out deterministically by sorting chip filenames alphanumerically within the five familiar regions (familiar_all[::7]) and selecting every seventh chip (USA: 10, India: 10, Paraguay: 10, Ghana: 7, Sri Lanka: 6). Because chips from the same flood event come from the same regional satellite overpass, ID test chips may exhibit regional spatial correlation with training chips; ID IoU is therefore used strictly as a familiar-domain reference, while primary generalization claims rely on the geographically disjoint OOD test set.
3. **Validation set (model selection, 72 chips):** composed of three independent events (Somalia 26, Pakistan 28, Nigeria 18), used exclusively for checkpoint selection and early stopping.
4. **Out-of-distribution (OOD) test set (unseen regions, 75 chips):** composed of three geographically distinct held-out events across three continents: Bolivia (15 chips; tropical Amazonian floodplain), Mekong (30 chips; Southeast Asian monsoonal delta) and Spain (30 chips; Mediterranean agricultural and semi-arid terrain).

3.2 SAR Preprocessing and Invalid-Pixel Handling

SAR segmentation is sensitive to the order of operations between invalid-pixel masking and numerical normalization. In raw_db, 0.0 dB represents an intense radar return; in minmax_clip, 0.0 represents the -30 dB floor; and in zscore, 0.0 represents the mean backscatter. To prevent invalid swath borders from distorting spatial filters or convolutional activations, each chip is processed in four sequential steps:

1. **Validity mask construction:** a binary validity mask $V \in \{0,1\}^{512\times512\times1}$ is defined prior to any transformation, as given in Eq. (1), where $B \in \{\mathrm{VV},\mathrm{VH}\}$.
2. **Leak-free training statistics:** global normalization parameters are computed strictly over valid pixels ($V_{i,j}=1$) in the familiar training regions: $\mu_{\mathrm{VV}} = -10.1397$ dB, $\sigma_{\mathrm{VV}} = 3.8141$ dB, 1st percentile $P_{01,\mathrm{VV}} = -22.50$ dB and 99th percentile $P_{99,\mathrm{VV}} = -1.50$ dB; for VH, $\mu_{\mathrm{VH}} = -16.9252$ dB, $\sigma_{\mathrm{VH}} = 4.3560$ dB, $P_{01,\mathrm{VH}} = -29.50$ dB and $P_{99,\mathrm{VH}} = -7.50$ dB.
3. **Pre-filter imputation (E8 only):** invalid pixels ($V_{i,j}=0$) are temporarily filled with the training mean μ_B before filtering, so that NaN values do not propagate through the 5×5 sliding window. This is imputation, not exclusion: the filled pixels enter the local mean and local variance of every window that overlaps them, so the filtered value of a valid pixel within two pixels of an invalid region is computed from a window that contains imputed values. The mask $V_{i,j}$ is still applied afterwards to all losses and metrics, and no imputed pixel is ever scored, but the boundary-adjacent valid pixels inherit a small bias toward μ_B. Because $\mu_{\mathrm{VV}} = -10.14$ dB sits inside the backscatter range rather than at an extreme, the imputed values are not outliers, which bounds the effect; it is nevertheless non-zero and is not removed by any later step. Masked local statistics (normalized convolution over V) would eliminate it and were not implemented.
4. **Post-normalization zero-clamping:** immediately after applying the chosen normalization formula (and after any intensity augmentation), all invalid pixels are clamped to $X_{i,j} = 0.0$. Because the first convolution layer is bias-free (use_bias=False), a zero input contributes nothing to that layer's output at the invalid pixel itself, and $V_{i,j}$ is passed directly to the loss and evaluation functions so that invalid pixels are excluded from all gradient terms and metrics. **This is a weaker guarantee than "no influence":** zero-clamping only neutralizes the invalid pixel's own input value. The same 3×3 kernel still reads the invalid pixel when it is centered on a neighboring valid pixel, so swath-border values propagate into the first feature map of adjacent valid pixels and, through four down-sampling stages, into receptive fields that extend well beyond the border. In addition, zero is a real value in every one of the normalizations used here (0.0 dB return in raw_db, the -30 dB floor in minmax_clip, the mean backscatter in zscore), so the clamped value is not a special token that the network can learn to ignore. Passing $V_{i,j}$ as an explicit third input channel, or replacing the border convolution with masked or partial convolutions, is the correct remedy and is left to future work (Section 5.4).

$$V_{i,j} = \begin{cases} 1, & \text{if } y_{i,j} \neq -1 \text{ and } B_{i,j} \notin \{\mathrm{NaN}, \pm\infty\} \text{ for } B \in \{\mathrm{VV},\mathrm{VH}\} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

3.2.1 Normalization Formulas:

Four input representations are evaluated for each band $B \in \{\mathrm{VV},\mathrm{VH}\}$:

1. **Raw dB (raw_db):** valid intensities remain in native decibel units, as in Eq. (2).

$$B_{\mathrm{raw}} = B \tag{2}$$

2. **Min-Max scaling (minmax_clip):** values are clipped to $[-30.0, 0.0]$ dB and scaled linearly to $[0,1]$, as in Eq. (3).

$$B_{\mathrm{mm}} = \frac{\min(\max(B,-30.0),0.0)+30.0}{30.0} \tag{3}$$

3. **Z-score standardization (zscore):** values are standardized to zero mean and unit variance, as in Eq. (4).

$$B_{\mathrm{z}} = \frac{B-\mu_B}{\sigma_B} \tag{4}$$

4. **Percentile clipping with standardization (percentile):** values are Winsorized to $[P_{01,B}, P_{99,B}]$ before Z-score standardization, as in Eq. (5).

$$B_{\text{pct}} = \frac{\min(\max(B, P_{01,B}), P_{99,B}) - \mu_B}{\sigma_B} \tag{5}$$

3.2.2 Lee-Style Speckle Filtering in the Log Domain:

In experiment E8, a 5×5 Lee-style local-statistics filter [8] is applied independently to the raw VV and VH bands in decibels, prior to Z-score standardization, using scipy.ndimage.uniform_filter with a 5×5 window. The local mean $\bar{I}$, local variance $\sigma^2_{\text{local}} = \max\left(\overline{I^2} - \bar{I}^2, 0\right)$ and chip-wide variance σ^2_{global} are all computed in the same decibel domain, and the filtered value $\hat{I}$ follows Eq. (6). Local statistics therefore include swath-border pixels filled with mu_B in the imputation step of Section 3.2, so valid pixels within two pixels of an invalid region are influenced by those imputed values (the magnitude is qualified in that step).

$$\hat{I} = \bar{I} + W\,(I - \bar{I}), \qquad W = \frac{\sigma^2_{\text{local}}}{\sigma^2_{\text{local}} + \sigma^2_{\text{global}} + \varepsilon} \tag{6}$$

where $\varepsilon = 10^{-8}$.

What the weight in Eq. (6) is, and is not: W is a signal-to-total variance ratio. It lies in $[0, 1]$, tends to 0 where the local variance is small relative to the chip-wide variance (homogeneous areas, where the estimate is replaced by the local mean) and to 1 where it is large (heterogeneous areas, where the observation is retained). This is the qualitative behavior expected of adaptive speckle filters, but it is not Lee's minimum-mean-square-error weight, which requires the speckle noise variance σ_n^2 (or, equivalently, the local coefficient of variation ratio $1 - C_u^2/C_I^2$) and not the chip-wide variance. Substituting σ^2_{global} for σ_n^2 makes Eq. (6) a heuristic variance-ratio smoother: it adapts to local heterogeneity but is not calibrated to the speckle statistics of the sensor.

Domain of application and its limits: the canonical Lee filter [8] is derived for linear intensity (or power), where speckle is multiplicative and the local coefficient of variation determines the filter weight. In the log domain, fully developed speckle becomes additive with an approximately signal-independent variance, which is the noise model that local-statistics filtering assumes, so applying the same adaptive weighting to decibel values is a defensible variant rather than an equivalent one. We therefore describe E8 as a Lee-style filter in the log domain and make no claim that it reproduces the canonical power-domain Lee filter; a linear-power implementation (filter before the dB conversion and after back-conversion) is a controlled comparison we did not run, and the E8 results in Sections 4 and 5 must be read with that restriction in mind.

3.3 Data Augmentation Regimes and Experimental Matrix

Four training augmentation regimes are implemented:

1. **No augmentation (none):** chips are fed in their native orientation and radiometric scale.
2. **Geometric augmentation (geometric):** random horizontal flips ($p = 0.5$), vertical flips ($p = 0.5$) and discrete 90° rotations ($k \in \{0, 1, 2, 3\}$ with equal probability 0.25) are applied identically to X, y and V.
3. **Intensity augmentation (intensity):** with probability $p = 0.5$ applied to the chip, a uniform calibration offset $\delta \sim \mathcal{U}(-0.15, 0.15)$ and additive zero-mean Gaussian noise $\eta \sim \mathcal{N}(0, 0.08^2)$ are added to all valid pixels ($V_{i,j} = 1$) in the normalized tensor X, followed by re-clamping of invalid pixels ($V_{i,j} = 0$) to 0.0. **Parameterization for reproducibility:** one scalar δ is drawn per chip and added to both polarization channels, modeling a common calibration drift, whereas η is drawn independently for every pixel and every channel (i.i.d. over the $512 \times 512 \times 2$ tensor), so the two bands receive the same offset but separate speckle-like perturbations.
4. **Combined augmentation (combined):** both geometric and intensity transformations are applied sequentially.

Training all 16 cells of the full factorial grid on a CPU workstation would require over 45 hours of compute, so we use a structured ten-experiment fractional design (Table 1). First, all four normalizations are evaluated with no augmentation (E0, E1, E2, E3) to isolate the main effect of normalization. Second, the augmentation regimes are evaluated under zscore against the E2 baseline: intensity (E5) and combined (E6) augmentation are fully controlled, whereas the geometric cell under zscore (E4_ZScore_Geometric) was trained on the 299-chip pilot split and is therefore exploratory rather than controlled (Section 4.2.1), so the main effect of augmentation is estimated from intensity and combined augmentation only. Third, two targeted interaction cells, MinMax_Geometric (minmax_clip × geometric) and E7_Percentile_Intensity (percentile × intensity), test whether spatial regularization rescues uncentered Min-Max scaling and whether outlier clipping stabilizes radiometric noise injection; these are supplemented by the Lee-style log-domain filter baseline (E8). The remaining seven cells (raw_db with geometric, intensity and combined augmentation; minmax_clip with intensity and combined; percentile with geometric and combined) were excluded to prioritize these primary physical hypotheses.

3.4 U-Net Architecture and Masked Loss Formulation

All experiments use an identical 2D U-Net [3] taking a $(512, 512, 2)$ input tensor. The encoder comprises four stages with 32, 64, 128 and 256 filters connected by 2×2 max-pooling layers, followed by a 512-filter bottleneck. Each convolutional block consists of two 3×3 bias-free convolutions (padding="same", use_bias=False), each followed by BatchNormalization [19] and a ReLU activation. The symmetric decoder upsamples feature maps via 2×2 nearest-neighbor interpolation, concatenates skip connections and applies twin 3×3 convolutional blocks. A 1×1 sigmoid convolution outputs water probabilities $\hat{y} \in (0, 1)^{512 \times 512 \times 1}$ (7,759,521 total parameters).

To handle class imbalance [20] and exclude invalid swath borders ($V_{i,j} = 0$), we optimize a custom masked binary cross-entropy (BCE) plus soft Dice loss, defined in Eqs. (7)-(9).

Table 1. Fractional experimental matrix of evaluated (✓) and excluded (×) configurations across normalization and augmentation regimes: nine of the sixteen factorial cells, plus one separate Lee-style filtering experiment (E8).

Configuration ID	Normalization	Augmentation	Status
E0_RawDB_NoAug	raw_db	none	✓ evaluated
E1_MinMax_NoAug	minmax_clip	none	✓ evaluated
E2_ZScore_NoAug	zscore	none	✓ evaluated
E3_Percentile_NoAug	percentile	none	✓ evaluated
E4_ZScore_Geometric	zscore	geometric	✓ evaluated
E5_ZScore_Intensity	zscore	intensity	✓ evaluated
E6_ZScore_Combined	zscore	combined	✓ evaluated
E7_Percentile_Intensity	percentile	intensity	✓ evaluated
E8_ZScore_LeeFilter[b]	zscore	none (5 × 5 Lee-style, dB)	✓ evaluated
MinMax_Geometric	minmax_clip	geometric	✓ evaluated
raw_db × geometric / intensity / combined	raw_db	geometric, intensity, combined	× excluded
minmax_clip × intensity / combined	minmax_clip	intensity, combined	× excluded
percentile × geometric / combined	percentile	geometric, combined	× excluded

Nine of the sixteen cells of the full normalization × augmentation factorial were trained (E0–E7 and MinMax_Geometric), which leaves the seven cells listed above untested; they were excluded to prioritize the primary physical hypotheses. [b]E8_ZScore_LeeFilter is not a cell of that factorial: it is a separate preprocessing experiment that inserts a 5×5 Lee-style log-domain filter into the Z-score pipeline, and it is listed here only for completeness.

$$\mathscr{L}_{\mathrm{BCE}} = -\frac{\sum_{i,j} V_{i,j}\left[y_{i,j}\ln\left(\hat{y}_{i,j}\right) + \left(1-y_{i,j}\right)\ln\left(1-\hat{y}_{i,j}\right)\right]}{\sum_{i,j} V_{i,j} + \varepsilon} \tag{7}$$

$$\mathscr{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_{i,j} V_{i,j}\, y_{i,j}\, \hat{y}_{i,j} + \varepsilon_d}{\sum_{i,j} V_{i,j}\left(y_{i,j} + \hat{y}_{i,j}\right) + \varepsilon_d} \tag{8}$$

$$\mathscr{L}_{\mathrm{total}} = \mathscr{L}_{\mathrm{BCE}} + \mathscr{L}_{\mathrm{Dice}} \tag{9}$$

where $\hat{y}_{i,j}$ is clipped to $[10^{-7}, 1-10^{-7}]$, $\varepsilon = 10^{-7}$ and $\varepsilon_d = 10^{-5}$.

3.5 Hyperparameters, Convergence and Evaluation Metrics

Models are implemented in Python 3.10 using TensorFlow/Keras 2.13, NumPy 1.24, SciPy 1.11 and Rasterio 1.3 on an Intel Core i5-7500T CPU workstation (2.70 GHz). Training uses an in-memory sequence generator (batch_size=2, 128 steps/epoch), the Adam optimizer [21] ($\mathrm{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), a fixed random seed (seed=42), a 10-epoch budget (≈ 3.2 hours/model), ModelCheckpoint saving the weights that maximize batch-averaged validation masked IoU (val_masked_iou, mode="max"), and EarlyStopping(monitor="val_masked_iou", mode="max", patience=5, restore_best_weights=True). Both callbacks therefore track the same quantity, so the weights restored at the end of training are the same best-val_masked_iou epoch that ModelCheckpoint saved, and every reported metric is computed from those restored weights. **Epoch coverage of the pilot pool:** the generator draws 128 batches of 2 chips per epoch, i.e., 256 chip presentations. For the eight controlled runs this equals exactly one pass over the 256-chip split. For the two pilot runs the pool is 299 chips, so one epoch consumes the first 256 entries of a permutation and leaves 43 undrawn. The index list is reshuffled at the end of every epoch (numpy.random.shuffle on the epoch-end hook) and the next epoch again draws the first 256 entries of the new permutation, so the 43 chips that were skipped move into the sampled block in later epochs; a fixed subset is never withheld. This also means that the number of distinct pilot chips actually seen depends on how many epochs a run completed before its best checkpoint (2 for E4_ZScore_Geometric, 6 for MinMax_Geometric), and that a given chip is not guaranteed to appear in a given epoch. As Table 2 shows, validation IoU peaked between epoch 2 and epoch 10 across all configurations (epoch 2 for E4; epoch 6 for E0, MinMax_Geometric, E3, E6 and E7; epoch 8 for E5 and E8; epoch 10 for E1 and E2), indicating that the selected validation checkpoints were reached within the 10-epoch training budget for the configurations evaluated.

During evaluation, probability maps are thresholded at $\hat{y}_{i,j} > 0.5$. Global pixel-level true positives (TP), false positives (FP) and false negatives (FN) are accumulated strictly over valid pixels ($V_{i,j} = 1$) to compute dataset-level Intersection over Union (IoU), Dice coefficient, precision, recall and the ID-to-OOD difference ($\Delta\mathrm{IoU}_{\mathrm{gap}}$), as defined in Eqs. (10)-(14).

$$\mathrm{IoU} = \frac{TP}{TP+FP+FN} \tag{10}$$

$$\mathrm{Dice} = \frac{2TP}{2TP+FP+FN} \tag{11}$$

$$\mathrm{Prec} = \frac{TP}{TP+FP} \tag{12}$$

$$\mathrm{Rec} = \frac{TP}{TP+FN} \tag{13}$$

$$\Delta\mathrm{IoU}_{\mathrm{gap}} = \mathrm{IoU}_{\mathrm{ID}} - \mathrm{IoU}_{\mathrm{OOD}} \tag{14}$$

To quantify dark-land shadow confusion in our backscatter analysis, we also define the exploratory dark-land overlap percentage ($\mathrm{DLO}_{<-15\,\mathrm{dB}}$) as the percentage of valid non-water land pixels ($y_{i,j} = 0$, $V_{i,j} = 1$) whose VV backscatter falls below the classical -15.0 dB water threshold [1], as given in Eq. (15),

Table 2. Detailed hyperparameter configuration and best validation checkpoint epochs across all ten experiments.

Exp. ID	Norm.	Lee-style (dB)	Aug. mode	Train chips	Best epoch	Peak val_masked_iou	Min val_loss
E0_RawDB_NoAug	raw_db	no	none	256	6	0.2650	0.9669
E1_MinMax_NoAug	minmax_clip	no	none	256	10	0.2458	1.0740
E2_ZScore_NoAug	zscore	no	none	256	10	0.2766	0.9512
E3_Percentile_NoAug	percentile	no	none	256	6	0.2695	0.9576
E4_ZScore_Geometric[a]	zscore	no	geometric	299[a]	2	0.2747	0.9685
MinMax_Geometric[a]	minmax_clip	no	geometric	299[a]	6	0.2531	1.0036
E5_ZScore_Intensity	zscore	no	intensity	256	8	0.2724	0.9401
E6_ZScore_Combined	zscore	no	combined	256	6	0.2756	1.2545
E7_Percentile_Intensity	percentile	no	intensity	256	6	0.2717	0.9796
E8_ZScore_LeeFilter	zscore	yes (5×5)	none	256	8	0.2666	0.9476

[a]Pilot runs. E4_ZScore_Geometric and MinMax_Geometric were trained on all 299 familiar-region chips before the 43-chip ID holdout was isolated, i.e., on 43 more chips than the other eight rows. Their OOD metrics on the 75 unseen test chips are leak-free, but their ID IoU is a training-set evaluation rather than a held-out result, and their OOD scores are not comparable with the 256-chip rows because the training sample size differs. They are therefore excluded from every controlled comparison in Sections 4 and 5 and are reported only as exploratory evidence (Section 4.2.1).

$$\mathrm{DLO}_{<-15\,\mathrm{dB}} = 100 \times \frac{N_{\mathrm{dark}}}{N_{\mathrm{land}}} \quad (15)$$

Here N_{land} is the number of valid non-water land pixels ($V_{i,j} = 1$, $y_{i,j} = 0$) and N_{dark} is the number of those pixels whose VV backscatter is below -15.0 dB.

4. Experimental Results

4.1 Quantitative Segmentation Matrix

Table 3 presents the complete quantitative evaluation. Its first eight rows are the controlled experiments, all trained on the identical 256-chip split and ordered by unseen OOD IoU; the two pilot runs are placed at the foot of the table, marked [a], and excluded from the ordering, so every comparative statement in this paper is made over the eight controlled rows only.

4.2 Normalization, Augmentation and Interaction Results

Every comparison in this subsection is restricted to the eight runs trained on the identical 256-chip split; the two 299-chip pilot runs are treated separately in Section 4.2.1. Among the four unaugmented normalization baselines (E0, E1, E2, E3), E0_RawDB_NoAug achieves the highest overall unseen generalization (OOD_IoU = 0.7837, OOD_Dice = 0.8788), the highest OOD precision (0.8965) and the highest accuracy in Spain (0.7174 IoU). E2_ZScore_NoAug achieves the highest held-out in-distribution accuracy (ID_IoU = 0.5131) and the highest Mekong IoU (0.8656). E3_Percentile_NoAug (OOD_IoU = 0.7760) improves over E2 in Bolivia by +1.99 percentage points (0.6955 vs. 0.6756) and in Spain by +0.37 percentage points (0.6988 vs. 0.6951). Conversely, the unaugmented E1_MinMax_NoAug performs worst across both familiar ID chips (0.4031 IoU) and unseen OOD events (0.6889 IoU, a decrease of 9.48 percentage points relative to E0).

Within the controlled set, no augmentation regime improved overall unseen accuracy: intensity augmentation on Z-score inputs (E5, 0.7614) is -1.57 percentage points below E2 (0.7771), combined augmentation (E6, 0.7311) is -4.60 below E2, Lee-style filtering (E8, 0.7447) is -3.24 below E2, and intensity augmentation on percentile-clipped inputs (E7, 0.7538) is -2.22 below E3 (0.7760). The losses are, however, regionally uneven in two distinct ways. E7, E8 and E6 all trade a Bolivia gain (up to +5.57 percentage points for E7 relative to E2, 0.6756 $\rightarrow$ 0.7313) for a larger Spain loss (-6.16 percentage points for E7, 0.6951 $\rightarrow$ 0.6335), and do so with higher OOD recall (0.9008–0.9276 vs. 0.8847) and lower OOD precision (0.7754–0.8146 vs. 0.8647). E5 is the exception: it lowers IoU in all three OOD regions (Bolivia 0.6390, Mekong 0.8541, Spain 0.6819) while leaving precision (0.8659) and recall (0.8632) close to the E2 baseline.

One interaction cell is fully controlled: percentile clipping $\times$ intensity augmentation, for which all four cells of the 2×2 design exist (E2 zscore/none, E3 percentile/none, E5 zscore/intensity, E7 percentile/intensity, all on the 256-chip split). We report it in two separate ways, because they answer different questions.

Direct configuration differences. Relative to its own unaugmented baseline, E7 gains +3.58 percentage points of Bolivia IoU (0.6955 $\rightarrow$ 0.7313) and loses -6.53 percentage points in Spain (0.6988 $\rightarrow$ 0.6335), raising OOD recall from 0.8625 to 0.9098 and lowering precision from 0.8855 to 0.8146. Relative to intensity augmentation on unclipped inputs, E7 scores +9.23 percentage points higher in Bolivia (0.6390 $\rightarrow$ 0.7313, +14.44% relative) and improves every OOD region. These are descriptive differences between individual configurations and should be quoted as such; they are not interaction effects.

Interaction contrast (difference-in-differences). The interaction itself is the change in the augmentation effect when normalization changes, $(E7 - E3) - (E5 - E2)$, computed separately for each region: $(0.7313 - 0.6955) - (0.6390 - 0.6756) = +0.0724$ for Bolivia (+7.24 percentage points), $(0.8570 - 0.8544) - (0.8541 - 0.8656) = +0.0141$ for Mekong (+1.41), $(0.6335 - 0.6988) - (0.6819 - 0.6951) = -0.0521$ for Spain (-5.21), and $(0.7538 - 0.7760) - (0.7614 - 0.7771) = -0.0065$ for overall OOD IoU (-0.65 percentage points). The sign therefore depends on the region: clipping makes intensity augmentation markedly

Table 3. Quantitative segmentation performance across familiar in-distribution (ID) and unseen out-of-distribution (OOD) flood events. The eight controlled runs, all trained on the identical 256-chip split, are sorted by OOD IoU; the two pilot runs marked [a] are listed separately at the foot of the table and are not part of the ranking.

Experiment	Norm.	Aug.	ID IoU	Bolivia	Mekong	Spain	OOD IoU	OOD Dice	OOD Prec	OOD Rec	ΔIoU
E0_RawDB_NoAug	raw_db	none	0.4809	0.6898	0.8563	0.7174	0.7837	0.8788	0.8965	0.8617	−0.3029
E2_ZScore_NoAug	zscore	none	0.5131	0.6756	0.8656	0.6951	0.7771	0.8746	0.8647	0.8847	−0.2640
E3_Percentile_NoAug	percentile	none	0.4797	0.6955	0.8544	0.6988	0.7760	0.8739	0.8855	0.8625	−0.2962
E5_ZScore_Intensity	zscore	intensity	0.5112	0.6390	0.8541	0.6819	0.7614	0.8645	0.8659	0.8632	−0.2502
E7_Percentile_Intensity	percentile	intensity	0.4587	0.7313	0.8570	0.6335	0.7538	0.8596	0.8146	0.9098	−0.2950
E8_ZScore_LeeFilter	zscore	Lee-style	0.4513	0.7226	0.8384	0.6343	0.7447	0.8536	0.8112	0.9008	−0.2934
E6_ZScore_Combined	zscore	combined	0.4534	0.7137	0.8535	0.5932	0.7311	0.8447	0.7754	0.9276	−0.2778
E1_MinMax_NoAug	minmax_clip	none	0.4031	0.6964	0.8080	0.5520	0.6889	0.8158	0.7539	0.8887	−0.2858
E4_ZScore_Geometric[a]	zscore	geometric	0.4288[a]	0.7273	0.8642	0.6269	0.7530	0.8591	0.8162	0.9067	−0.3241
MinMax_Geometric[a]	minmax_clip	geometric	0.4371[a]	0.7015	0.8566	0.6122	0.7400	0.8506	0.8033	0.9038	−0.3029

[a]Pilot runs trained on the 299-chip familiar pool (43 chips more than every other row) before the ID holdout was isolated. Their ID IoU is a training-set evaluation, not a held-out result, and their OOD scores are confounded with the larger training sample; all comparisons in this paper are therefore restricted to the eight 256-chip rows.

less harmful in Bolivia, does little in Mekong, and makes it more harmful in Spain, with a slightly negative net effect on overall OOD IoU. On this evidence percentile clipping does not generally improve the effect of intensity augmentation; it redistributes it between regions. All four contrasts are single-seed and descriptive, with no p-values or confidence intervals attached (Section 4.3). Fig. 1(b) shows the cross-regional profile of the controlled runs.

4.2.1 Pilot Runs on the 299-Chip Split: Exploratory Evidence:

Two further runs, E4_ZScore_Geometric and MinMax_Geometric, were trained before the 43-chip in-distribution holdout was separated and therefore used 299 training chips, 16.8% more than every other row in Table 3. For both, the augmentation state and the training sample size differ from the 256-chip baseline they would be compared against, so no part of their difference can be assigned to augmentation. We report them only because they motivate a testable hypothesis, and we state explicitly what that test requires.

Against the 256-chip baselines, the pilot pair separates in opposite directions depending on normalization: MinMax_Geometric (0.7400) lies +5.11 percentage points above E1_MinMax_NoAug (0.6889), whereas E4_ZScore_Geometric (0.7530) lies −2.41 percentage points below E2_ZScore_NoAug (0.7771), its loss concentrated in Spain (0.6951 → 0.6269, −6.82 percentage points) with higher recall (0.9067 vs. 0.8847) and lower precision (0.8162 vs. 0.8647). Both pilot runs also carry a non-held-out ID IoU (0.4288 and 0.4371 from Table 3), which must not be read as in-distribution generalization. Fig. 1(a) displays this pair.

This pattern is consistent with a normalization × geometric augmentation crossover, but with one pilot run per cell, 43 extra training chips in both, and a single training seed, it is recorded here as a hypothesis rather than a result. The comparison needed to test it is specified in Section 5.4.

4.3 Effect Sizes, Inference Limits and Convergence Dynamics

All differences reported above are descriptive effect sizes between single-seed runs, and we deliberately attach no p-values or confidence intervals to them. Two features of the data make the usual tests inapplicable. First, the three OOD test events are not exchangeable samples of a regional population; they were fixed by the benchmark, so a test across them (for example a Friedman test over three events) has three observations and cannot support a general claim about regions. Second, satellite pixels are strongly spatially autocorrelated: the 75 unseen OOD chips contain approximately 2.0×10^6 union water pixels, but the effective number of independent observations is closer to the number of chips or flood events. Treating those pixels as independent samples would shrink the standard error by several orders of magnitude and would make differences of 0.001 IoU appear decisive, which is not a defensible inference for this design. A paired bootstrap that resamples whole chips (or flood events) across several random training seeds is the appropriate test, and the single-seed protocol prevents us from computing it; we therefore report raw differences and defer any claim of statistical significance.

What the regional values do show descriptively is that Mekong is the best-performing of the three unseen events in every one of the eight controlled runs (mean IoU 0.8484, range 0.8080–0.8656), whereas Bolivia (mean 0.6955, range 0.6390–0.7313) and Spain (mean 0.6508, range 0.5520–0.7174) exchange places: Spain is the higher of the two in E0, E2, E3 and E5, and Bolivia is the higher in E1, E6, E7 and E8. That exchange is itself the main regional effect reported above, and it is a description of these three benchmark events rather than a general claim about regions.

Fig. 3 shows the epoch-by-epoch training loss and validation IoU trajectories logged during training. While standardized models (E3, E7, E8) reach near-peak validation IoU by epochs 3–6, E1_MinMax_NoAug has a delayed validation warm-up (val_masked_iou = 0.0255 at epoch 1 versus 0.2022 for E7) and plateaus at a lower validation ceiling (0.2458 at epoch 10).

4.4 Regional Backscatter and Contrast Measurements

Table 4 reports the physical VV and VH backscatter means, water-to-land contrasts (Δ dB = $\mu_{\text{land}} - \mu_{\text{water}}$) and exploratory dark-land overlap ($\text{DLO}_{<-15\,\text{dB}}$) across all 11 Sen1Floods11 events. Two definitions apply throughout the table. Water % is defined as 100×(valid water pixels)/(valid pixels), so the denominator is the masked set $V_{i,j} = 1$ and swath-border pixels are excluded from numerator and denominator alike; it is not a fraction of all raster pixels. The backscatter means are pooled over all valid pixels

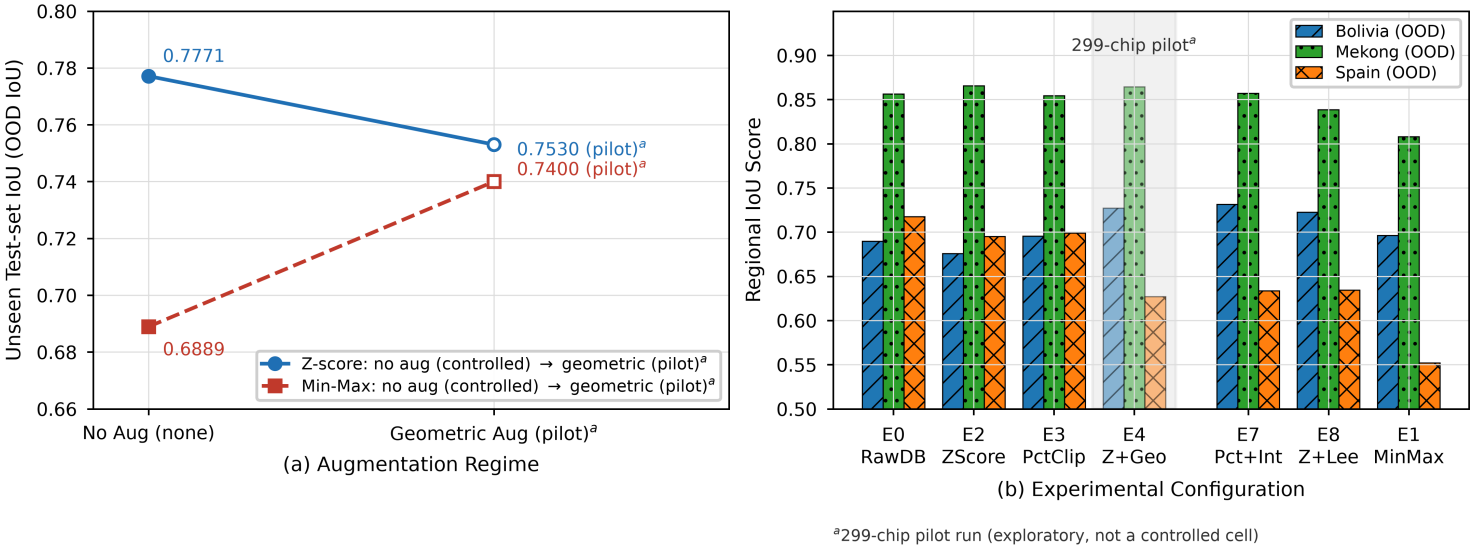


Fig. 1. (a) Pilot comparison of SAR normalization (zscore versus minmax_clip) with and without geometric augmentation, on unseen OOD IoU; both augmented runs were trained on 299 chips rather than 256, so this panel is exploratory (Section 4.2.1). (b) Regional IoU for Bolivia, Mekong and Spain across seven configurations: solid bars are the six controlled runs and the faded, hatched bar in the shaded band is E4, the 299-chip pilot run, which is shown for completeness and is not a controlled cell of the factorial.

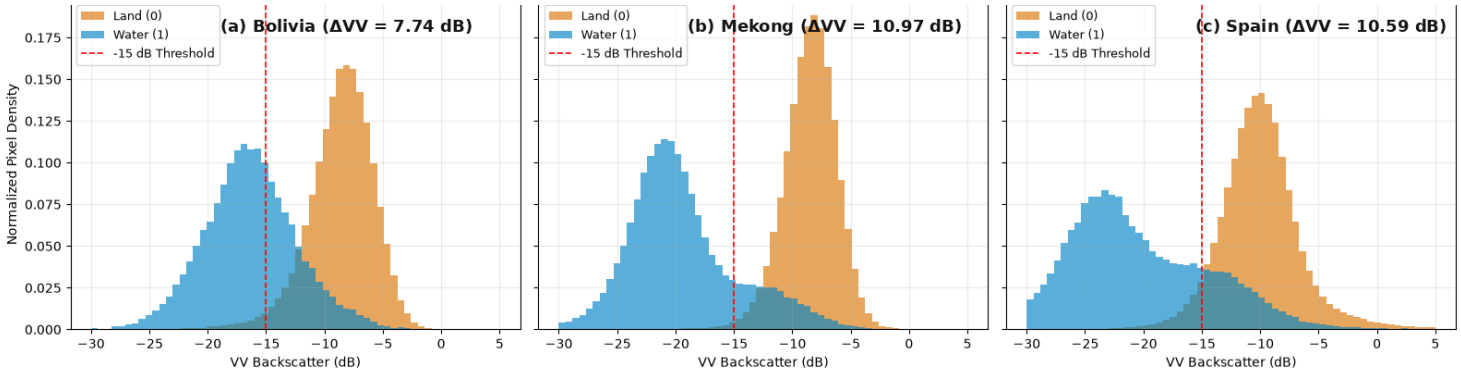


Fig. 2. Normalized pixel-density histograms of Sentinel-1 VV backscatter (dB) for non-water land (0) and surface water (1) across the three unseen OOD test regions: (a) Bolivia (ΔVV = 7.74 dB), (b) Mekong (ΔVV = 10.97 dB) and (c) Spain (ΔVV = 10.59 dB), with the exploratory −15 dB shadow-overlap threshold indicated by the dashed vertical line.

of an event rather than averaged from per-chip means, so events contribute in proportion to their valid area. Because validity comes only from the label mask and finite VV/VH values (Section 3.1), both quantities inherit any NoData convention those rasters encode without setting the label to −1.

5. Discussion

Section 4 reports what was measured; this section offers explanations for it. The two are kept separate throughout: quantitative statements about IoU, precision, recall and backscatter contrast are measurements over the chips described in Section 3, whereas mechanism statements are proposed explanations that the present experiments constrain but do not test directly. No mechanism below is claimed to have been demonstrated, and no per-pixel attribution of individual false positives was performed.

5.1 Candidate Mechanisms of Normalization and Augmentation Interaction

Measured: preprocessing and augmentation choices do not behave independently. Under minmax_clip ($[-30, 0]$ dB mapped to $[0, 1]$), the unaugmented run (E1_MinMax_NoAug) has the lowest accuracy in the study, including the lowest Spain IoU (0.5520), and Fig. 3 shows that it also warms up more slowly in early validation than the zero-centered runs. Among the controlled runs, every augmentation regime lowered overall OOD IoU while redistributing errors between Bolivia and Spain (Section 4.2).

Proposed explanations, none of which is isolated by these experiments. For the min-max deficit, one candidate is the interaction between the $[0, 1]$ input range and the post-normalization zero-clamping of invalid border pixels; another is the larger effective learning rate implied by non-centered inputs. Our design varies neither, so the two cannot be separated here. For the Bolivia/Spain

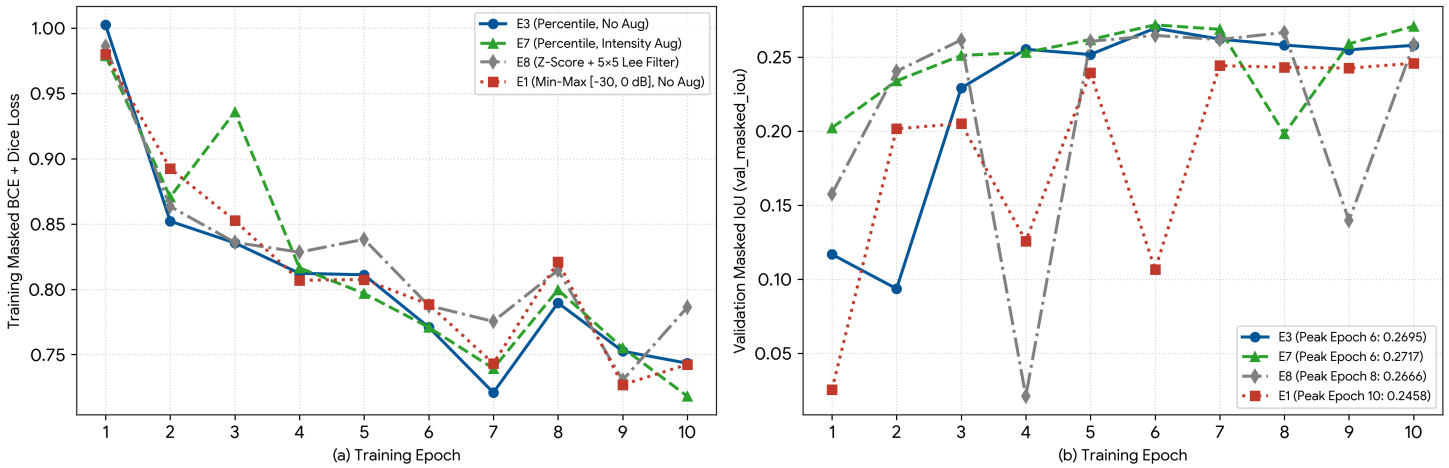


Fig. 3. Training loss and validation masked IoU convergence trajectories across the 10-epoch training budget for representative configurations (E1, E3, E7 and E8), illustrating slower early validation warm-up for non-zero-centered Min-Max scaling (E1) compared to standardized inputs.

Table 4. Physical radar backscatter statistics, water-to-land contrast (Δ dB) and dark-land overlap ($\mathrm{DLO}_{<-15\,\mathrm{dB}}$) across all 11 Sen1Floods11 events. Water % and all means use valid pixels only ($V_{i,j} = 1$) and are pooled per event.

Region	Split	Chips	Water %	VV water (dB)	VV land (dB)	VV Δ (dB)	VH water (dB)	VH land (dB)	VH Δ (dB)	$\mathrm{DLO}_{<-15\,\mathrm{dB}}$ (%)
Bolivia	OOD test	15	16.00	−16.37	−8.62	7.74	−24.47	−14.85	9.62	2.44
Mekong	OOD test	30	24.12	−19.46	−8.48	10.97	−28.32	−14.94	13.38	0.84
Spain	OOD test	30	13.15	−20.57	−9.97	10.59	−29.27	−17.08	12.19	5.67
Ghana	Train / ID	53	6.29	−15.20	−9.38	5.83	−23.38	−15.75	7.63	1.41
India	Train / ID	68	12.37	−16.78	−8.33	8.45	−24.17	−15.12	9.05	2.19
Paraguay	Train / ID	67	8.11	−16.12	−9.75	6.37	−25.86	−16.76	9.10	3.01
Sri Lanka	Train / ID	42	11.80	−13.12	−8.91	4.21	−24.44	−15.33	9.10	2.95
USA	Train / ID	69	4.56	−17.72	−10.98	6.74	−26.75	−17.19	9.56	10.08
Nigeria	Validation	18	23.56	−17.32	−9.92	7.40	−25.27	−16.06	9.21	2.48
Pakistan	Validation	28	7.50	−14.13	−10.11	4.02	−23.16	−18.18	4.99	9.40
Somalia	Validation	26	6.51	−15.02	−12.04	2.98	−22.10	−18.37	3.74	16.27

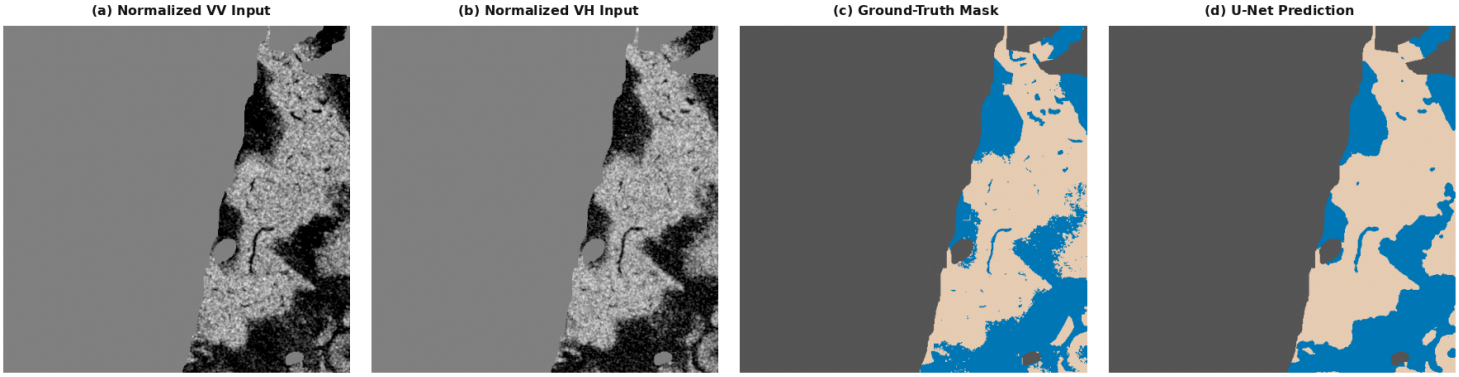


Fig. 4. Qualitative segmentation example on unseen test chip Bolivia_103757_S1Hand.tif: (a) Z-score normalized VV input, (b) Z-score normalized VH input, (c) expert hand-labeled ground-truth mask and (d) U-Net predicted water mask (gray denotes invalid swath borders excluded via validity mask V, tan denotes non-water land and blue denotes surface water).

exchange, one candidate is the directional structure of side-looking radar geometry: because topographic foreshortening (bright radar-facing slopes) and radar shadows (dark trailing slopes) occur in a consistent sequence relative to the look direction within an overpass, 90° rotations and 180° flips applied in pixel coordinates may weaken the association between a dark patch and its slope con-

text. This is a hypothesis; testing it requires ablating individual geometric transforms, which we did not do.

Augmentation is applied in normalized units, not in decibels. The intensity augmentation perturbs the normalized tensor X with an offset $\delta \sim \mathcal{U}(-0.15, 0.15)$ and noise $\eta \sim \mathcal{N}(0, 0.08^2)$; these are dimensionless quantities whose physical size depends on the normalization in force. Using the training statistics of Section 3.2.1 ($\sigma_{VV} = 3.8141$ dB, and a 30 dB span mapped onto $[0,1]$ for minmax_clip), one normalized unit corresponds to about 3.81 dB under Z-score or percentile standardization and about 30 dB under min-max scaling, so the same nominal perturbation corresponds to roughly 0.31 dB versus 2.4 dB of noise (about an eight-fold difference). These figures are approximate scale conversions computed from the stated normalization ranges and pooled band statistics, not exact per-pixel physical equivalents: the local conversion factor varies across an image and between the VV and VH bands, so they indicate the order of the discrepancy rather than a precise decibel value at any given pixel. We therefore make no statement that any augmented input lies in a particular physical decibel range, and comparisons of the augmentation magnitude across normalizations are not physically interchangeable. A candidate explanation for the Bolivia behavior of E7 is that clipping to $[P_{01}, P_{99}]$ bounds the input tails, so an injected offset or noise draw cannot move a floodplain-boundary pixel as far into the land-like range as it can on unclipped Z-score inputs; this is consistent with the measured +9.23 percentage-point Bolivia advantage of E7 over E5 (a direct configuration difference; the difference-in-differences contrast is $+7.24$ points there and -0.65 overall), but we did not measure the per-pixel displacement produced by augmentation, so the mechanism remains unverified.

5.2 Regional Backscatter Contrast and Candidate Physical Explanations

Table 4 and Fig. 2 give the measured backscatter statistics of the three unseen events and their candidate physical correlates. The associations are descriptive: the contrast measures are computed over whole events and are not linked pixel by pixel to the model errors, so they can support but cannot establish a cause for the regional ordering.

1. **Mekong (IoU = 0.8080–0.8656):** Mekong has the largest physical water-to-land separation in the benchmark (ΔVV = 10.97 dB, ΔVH = 13.38 dB) and the lowest dark-land overlap ($DLO_{<-15\,dB} = 0.84\%$), and its water and land histograms are the most clearly bimodal of the three (Fig. 2(b)). Segmentation accuracy exceeds 0.80 for every controlled run, which is consistent with, but does not by itself establish, radiometric separation being the dominant factor.
2. **Spain (IoU = 0.5520–0.7174):** although Spain has a wide mean separation (ΔVV = 10.59 dB), 5.67% of its valid non-water land pixels fall below -15.0 dB, nearly seven times the rate in Mekong (0.84%), and in Fig. 2(c) the Spanish land histogram carries a low-backscatter tail into the water regime (-15 to -25 dB). The recall-oriented runs (E6, E7, E8) lose 6.08–10.19 percentage points of Spain IoU relative to their baselines while gaining recall, and the precision-oriented unaugmented runs (E0, E3) keep the highest precision. These two measurements are consistent with part of the dark-land tail being labeled water, but they do not demonstrate it: smooth wet agricultural soils and topographic shadows are candidate physical causes of that tail that we did not verify, and confirming the attribution would require per-pixel error analysis against auxiliary slope or land-cover data, which was not performed.
3. **Bolivia (IoU = 0.6390–0.7313):** Bolivia has the narrowest VV contrast of the three unseen events (ΔVV = 7.74 dB) with mean water backscatter at -16.37 dB (Fig. 2(a)). Measured: the two lowest OOD recall values belong to unaugmented runs (E0 0.8617, E3 0.8625) and the two highest to recall-oriented runs (E6 0.9276, E7 0.9098), and E7 reaches the best Bolivia IoU (0.7313). Candidate explanation, not verified here: partially submerged floodplain vegetation raises water backscatter toward the land range, so a decision boundary tuned for high precision under-segments such pixels; confirming this would require inspecting vegetation masks at the error locations, which we did not do (Fig. 4 shows one qualitative example only).

5.3 Explaining the Negative ID-to-OOD Generalization Gap

Across the eight controlled runs in Table 3, familiar-region ID_IoU (0.4031–0.5131) is numerically lower than unseen OOD_IoU (0.6889–0.7837), yielding a negative ΔIoU_{gap} (-0.2502 to -0.2962). The network therefore does not generalize better to unfamiliar countries than to its own held-out in-distribution chips. Table 4 is consistent with the inversion arising from intrinsic event difficulty and foreground class prevalence, which is an explanation rather than a demonstration:

- First, the five familiar regions contain sparse, fragmented water bodies (USA: 4.56% water, Ghana: 6.29%, Paraguay: 8.11%), where small boundary false positives heavily penalize the IoU denominator ($TP + FP + FN$). In contrast, the unseen OOD test set is dominated by large, contiguous riverine floods in Mekong (24.12% water) and Bolivia (16.00% water).
- Second, the familiar regions exhibit much narrower physical VV water-to-land contrast (Sri Lanka: 4.21 dB, Ghana: 5.83 dB, Paraguay: 6.37 dB, USA: 6.74 dB) and high dark-land overlap in the USA ($DLO_{<-15\,dB}$ = 10.08%) compared to Mekong (ΔVV = 10.97 dB, $DLO_{<-15\,dB}$ = 0.84%).
- Third, the validation regions (Somalia: ΔVV = 2.98 dB, $DLO_{<-15\,dB}$ = 16.27%; Pakistan: ΔVV = 4.02 dB) represent the most radiometrically ambiguous scenes in Sen1Floods11, which is consistent with batch-averaged validation IoU peaking around 0.2458–0.2766 while the selected checkpoints still reach >0.68 IoU on unseen OOD events.

5.4 Limitations and Future Work

The results have several limitations:

1. **Single-seed training protocol:** owing to CPU hardware constraints ($\approx$ 32 hours of total compute across the ten models), each configuration was trained once with a fixed deterministic seed (seed=42) for 10 epochs. Single-seed runs cannot separate small model-to-model differences (such as the 0.0066 OOD

IoU gap between E0 and E2) from weight-initialization variance, and the large pixel count does not compensate: spatially autocorrelated pixels are not independent observations. We therefore report no p-values or confidence intervals anywhere in this paper, and none of the differences discussed should be read as statistically significant. The appropriate analysis, not available under the present protocol, is a paired bootstrap that resamples whole test chips (or the three flood events) across 3–5 random training seeds per configuration and reports mean $\pm$ standard deviation.

2. **Incomplete factorial grid and ablation granularity:** seven of the 16 possible normalization $\times$ augmentation combinations were omitted, and geometric augmentation bundled horizontal flips, vertical flips and 90° rotations together. Future ablations should isolate individual geometric transforms (e.g., flips vs. 90° rotations) to test the side-looking radar shadow hypothesis directly.
3. **Invalid-pixel handling at swath borders:** two effects remain. First, setting invalid border pixels to 0.0 after normalization makes the border value mean different physical things in raw_db, minmax_clip and zscore, and because the first convolution also reads those pixels from neighboring valid positions, border values reach feature maps and later layers despite the masked loss. Second, the Lee-style filter of E8 imputes invalid pixels with μ_B rather than excluding them, so imputed values enter the local statistics of adjacent valid pixels. Supplying the validity mask V as an explicit third input channel, using partial or masked convolutions, and computing mask-weighted local statistics would remove both effects; none of the three was implemented here.
4. **NoData provenance in the GeoTIFF masks:** the validity mask of Eq. (1) uses the label value -1 and a finiteness test on VV/VH only. rasterio's NoData metadata (src.nodata, src.nodatavals) and src.dataset_mask() were not consulted, and no finite NoData sentinel was searched for, so any raster that encoded invalid areas through those mechanisms would have contributed to the training statistics, the backscatter means and the water fractions of Table 4 rather than being excluded. Sen1Floods11 documents its hand-labeled masks as using -1 for invalid pixels, which is what Eq. (1) implements, but the rasters themselves were not audited against their own NoData metadata. A re-audit that compares src.dataset_mask() and src.nodata against the label mask, and re-derives μ_B, σ_B, P_{01} and P_{99} from the intersection, is the recommended next check.
5. **Pilot split difference:** E4_ZScore_Geometric and MinMax_Geometric were trained on all 299 familiar chips, i.e., 43 more chips than the eight controlled runs, before the ID holdout was isolated. Their unseen OOD metrics are leak-free but their training sample differs, so they are excluded from every controlled comparison and are reported only as the exploratory observation of Section 4.2.1. The comparison needed to settle the minmax_clip $\times$ geometric and zscore $\times$ geometric cells is explicit: retrain both runs on the identical 256-chip split with the same seed, epoch budget and schedule, then re-evaluate on the same 43 ID chips and 75 OOD chips and repeat across 3–5 seeds.
6. **Lee-style filter in the log domain:** the filter of Section 3.2.2 operates on decibel values, whereas the canonical Lee formulation [8] applies to linear intensity or power. The two are not equivalent, and we did not run the linear-power variant, so the E8 result characterizes a log-domain local-statistics filter and not the canonical filter; a power-domain re-implementation is the obvious next control.
7. **Augmentation magnitude is normalization-dependent:** as set out in Section 5.1, the offsets and noise added during intensity augmentation are expressed in normalized units, so their physical magnitude differs by roughly a factor of eight between Z-score/percentile and min-max inputs. Cross-normalization statements about the strength of the intensity augmentation are therefore not physically like-for-like.

6. Conclusion

This study examined how SAR normalization strategies and data augmentation regimes interact under cross-regional surface-water segmentation, using ten U-Net runs on 446 hand-labeled Sentinel-1 chips from Sen1Floods11, eight of which share an identical 256-chip training split. Over those eight controlled runs, three results stand. First, retaining native decibel values without augmentation gave the highest unseen OOD IoU (0.7837), ahead of Z-score (0.7771) and percentile (0.7760) standardization and well ahead of Min-Max scaling (0.6889, a 9.48 percentage-point deficit). Second, every controlled augmentation or filtering configuration, namely radiometric intensity, combined augmentation and Lee-style log-domain filtering, lowered overall OOD IoU relative to its own normalized baseline, but redistributed errors between regions: the recall-oriented configurations gained up to 5.57 percentage points of Bolivia IoU and lost 6.08–10.19 percentage points of Spain IoU as OOD recall rose to 0.9008–0.9276. Third, percentile clipping combined with intensity augmentation (E7) reached the best Bolivia IoU of the study (0.7313) and the highest controlled OOD recall (0.9098), although its overall OOD IoU (0.7538) remained below the unclipped baseline. Its difference-in-differences contrast against the Z-score pair is $+7.24$ percentage points in Bolivia but -0.65 overall and -5.21 in Spain, so clipping changes where the augmentation effect falls rather than improving it in general. Two pilot runs trained on 43 additional chips point to a normalization $\times$ geometric crossover (Min-Max +5.11, Z-score -2.41 percentage points), which we report as an untested hypothesis and not as a finding. Across all 11 benchmark events the backscatter statistics are consistent with regional accuracy depending on VV/VH water-to-land contrast and on dark-land overlap below -15 dB, but those associations are descriptive and do not attribute individual errors to agricultural land or radar shadow. Because all runs are single-seed, no statistical significance is claimed. The immediate priorities follow directly: retrain the two 299-chip pilot cells on the identical split, run a linear-power Lee control, express the intensity augmentation in physical units per normalization, and report paired chip-level bootstraps over 3–5 seeds.

Declarations

Funding: This research did not receive any specific grant or financial support from funding agencies in the public, commercial or

not-for-profit sectors.

Competing interests: The author declares no financial or non-financial competing interests or conflicts of interest relevant to this article.

Ethics approval and consent to participate: Not applicable. This study analyzes publicly available satellite remote sensing imagery and does not involve human participants or animal subjects.

Data availability: The Sen1Floods11 (v1.1) dataset analyzed in this study is publicly available from the original authors [1] at https://github.com/cloudtostreet/Sen1Floods11. The complete source code, trained model checkpoints, configuration files and geographical split lists supporting the findings of this study are openly available on GitHub at https://github.com/Sandeepssh/FloodShift.

Author contributions: Sandeep Hosamani conceived and designed the study, implemented the preprocessing and U-Net training pipeline, executed all experiments and statistical radar-contrast analyses, and wrote and edited the manuscript.

Disclosure of AI assistance: In accordance with journal policy on artificial intelligence, the author discloses that an AI language model was used for coding assistance, structuring analysis scripts and language editing. All local model training, data verification, scientific interpretation and manuscript review were performed by the author, who assumes full responsibility for the accuracy and integrity of the work.

Acknowledgement

The author thanks the creators of the Sen1Floods11 dataset and the European Space Agency (ESA) Copernicus Sentinel-1 mission for providing open access to hand-labeled SAR flood imagery.

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